

2.1 The History of the Operating System

The earliest computers were built for a particular purpose, and had their programming hard-wired into them. Data was given to the program through punch-cards, which could be altered easily. But to change the purpose of the machine, you needed to physically alter it, or buy a new one altogether.

Von Neumann to the Rescue

The Hungarian scientist John Louis von Neumann was the first to propose the idea of the “stored program”—where the code for the program was in the same memory area as the data. He figured (and correctly, too) that this wouldn’t be a problem, because the computer could tell the difference between data and code.

Putting data and code together meant that computers could be more flexible, and reprogramming them wouldn’t be such a headache anymore.

Getting your work done

This is how you would write and run your program in the good old days:

- m First, figure out what it is you want to write.
- m Punch out your program and data on a paper tape or punch-card.
- m Wait in a queue of programmers outside the computer room.
- m Enter your program into the computer.
- m Wait for your program to finish running or crash.
- m Repeat if desired.

This would become quite tedious, and soon it was proposed that some scheduling needed to be done. Programmers were given appointments in the day to come and run their programs. At the end of their appointment, they would give the machine to the next programmer. As the computers got more powerful, the time taken to finish a program grew lesser, and the computer would sit idle for longer. This was unacceptable—these were expensive machines, and people could not afford to let it just *lie* there, could they?